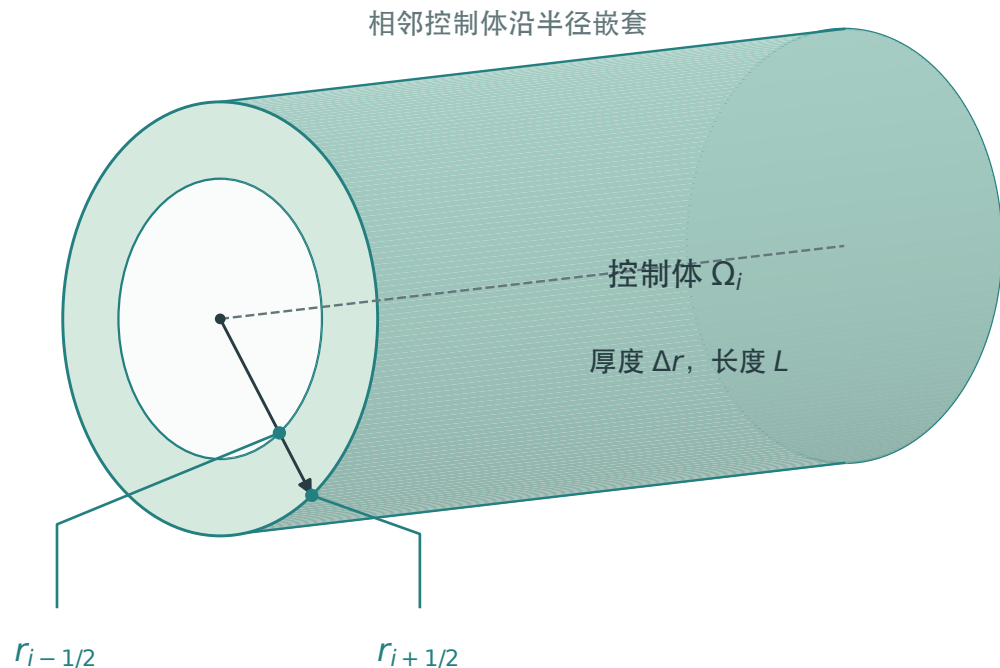


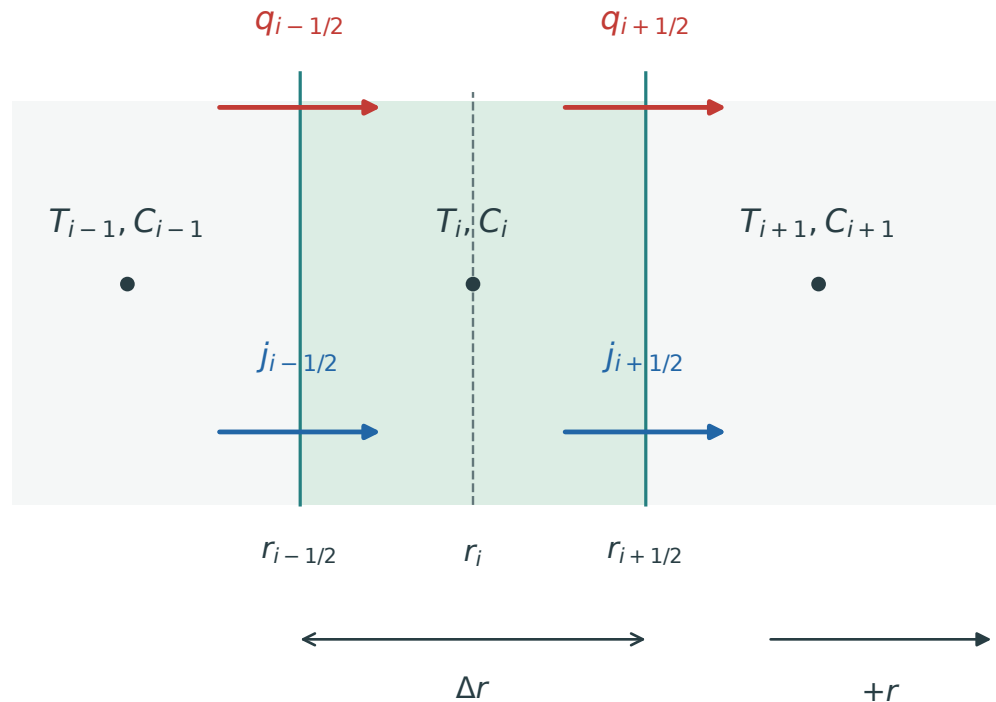
(a) 三维环形控制体



实际体积: $|\Omega_i| = 2\pi L V_i$

几何权重: $V_i = (r_{i+1/2}^2 - r_{i-1/2}^2)/2$

(b) 径向局部展开与共享通量



箭头表示通量正方向; 同一界面在相邻单元中异号计入

中心界面 $r_{1/2} = 0$, 面积加权通量为零